BB 101 BIOLOGY · END-SEMESTER EXAMINATION

Practice Questions & Answers

Lectures 15–23: Mendelian genetics, cell division, chromosomes & inheritance, cell communication, big data in biology, evolution and population genetics.

QUESTIONS
225 in total, 9 formats

REFERENCE
Campbell Biology, 10th ed. — Ch. 11–15, 16.1, 21.2, 22–24

PREPARED FOR
Tutorial Batch T1

Contents

1 Exam pattern & strategy — what to expect and how to attack each question type

2 Model question paper — 60 marks, 90 minutes (attempt it under exam conditions)

3 Topic-wise practice — Sections A to H, every lecture of the end-sem syllabus

4 Answers & explanations — with the Campbell section for each question

HOW TO USE THIS BOOKLET

Do the model paper first, timed, without notes. Mark it with Part 4, then go to the topic sections where you lost marks. Every answer gives the reason, not just the letter — read the explanation even when you got the question right.

Questions are written from the end-sem lecture slides and Campbell Biology (10th ed.). Section numbers such as “Campbell 14.1” point to the matching concept in the textbook. Items marked “—” are covered in the lecture slides only.

1 · Exam pattern and strategy

What the end-sem covers, the question formats, and the high-yield question patterns

1.1 Syllabus weightage

The end-sem syllabus is the 255-slide lecture PDF (Lectures 15–23). The share of slides is a rough guide to how much each topic matters. Lecture 16 largely repeats Lecture 15, so Section A's share is slightly inflated.

Sec.	Topic	Slides	Share	Typical questions	Qs here
A	Heredity and Mendel's laws Lectures 15–16 · Campbell Ch. 14.1	58	23%	Ratios (3:1, 1:2:1, 9:3:3:1), terminology, history, 'why 3:1?'	23
B	Mendel and the gene idea: probability, dominance, pedigrees Lecture 17 · Campbell Ch. 14.1–14.4	24	9%	Probability numericals, ABO, dominance types, pedigrees	23
C	Cell cycle, mitosis and meiosis Lecture 18 · Campbell Ch. 12 and 13	23	9%	Stage → event matching, chromosome/DNA counting, mitosis vs meiosis	26
D	Chromosomal basis of inheritance Lecture 19 · Campbell Ch. 15	17	7%	X-linked crosses, Morgan, nondisjunction, syndromes	23
E	Cell communication and signal transduction Lecture 20 · Campbell Ch. 11	49	19%	Signalling types, GPCR/G-protein steps, cAMP pathway, amplification	25
F	Big data in biology and bioinformatics Lecture 21 · Lecture slides; Campbell 16.1, 21.2	29	11%	-omes, prefixes, DNA forms, definitions, AlphaFold	24
G	Evolution and natural selection Lecture 22 · Campbell Ch. 22, 23 (opening), 24.1	35	14%	Darwin, finches, species concept, MRSA, observations/inferences	21
H	Population genetics and Hardy–Weinberg Lectures 22–23 · Campbell Ch. 23	20	8%	Hardy–Weinberg numericals, drift/gene flow, selection types	24

READING THE TABLE

Mendel (A–B), cell communication (E) and evolution with population genetics (G–H) together make up nearly three-quarters of the slides — give them most of your revision time. The numerical questions come mainly from B, D and H.

1.2 Question formats and how to attack them

Format	What it really tests	Strategy
MCQ — single correct	Facts, definitions, 'which is NOT…'	Read the stem twice; underline NOT / EXCEPT. Eliminate the two clearly wrong options first.
One or more correct	Whether you know every statement	Judge each option as its own True/False. Do not assume at least two are right.

Match the following	Pairs of terms, stages, scientists, examples	Start with the pairs you are sure of; Column II may contain an extra (distractor) item.
Fill in the blanks	Exact terms (often from a figure)	Spell the term exactly as in the lecture: *prometaphase*, *hemizygous*, *plasmodesmata*.
True / False	Common misconceptions	One wrong word makes the whole statement false. For 'false', write the correction.
Assertion – Reason	Cause and effect	Decide A and R separately first; only then ask "does R explain A?".
Numerical	Crosses, probability, Hardy–Weinberg, counting	Write genotypes and gametes first; one gene at a time; check that fractions sum to 1.
Short answer / figure	Explaining a mechanism; reading a pedigree or micrograph	3 marks ≈ 3 points. Use the lecture's key words (e.g. reception, transduction, response).

This booklet contains: MCQ 102, Numerical 30, Match the following 19, True / False 19, Short answer 18, Fill in the blanks 14, Assertion – Reason 12, One or more correct 9, Arrange in order 2.

1.3 High-yield question patterns

Pattern	What to expect	Practise with
Figure-based fill-in and labelling	The tutorials used blank-filling on lecture figures (Mitosis and Mendel worksheets). Expect labelled diagrams: mitosis stages, Punnett squares, pedigrees, the GPCR cycle.	C13, C16, B14, E18
Ratios and probability	3 : 1, 1 : 2 : 1, 9 : 3 : 3 : 1, the trihybrid 'at least two recessive' problem from the slides, ABO and albinism crosses.	A19–A21, B15–B20
X-linked crosses	The three colour-blindness crosses (slide 118) and Morgan's fruit flies are the core of Lecture 19 — expect at least one cross.	D13–D17
Hardy–Weinberg numericals	Two worked examples were added to the end of the slides (552 pea plants; lizard scales). Know both routes: from q^2 and by counting alleles.	H13–H19
Compare and contrast	Mitosis vs meiosis, natural vs artificial selection, explainable vs interpretable AI, paracrine vs endocrine, the three modes of selection.	C25, G19, F7, E16, H11
Who / when	Mendel 1866, Griffith 1928, Avery 1944, Watson & Crick 1953, Darwin 1859, Sutherland 1971, Gilman & Rodbell 1994, AlphaFold Nobel 2024, Dayhoff ~1960.	A9, E13, F3, F10
'Why' questions from the slides	"Why is the ratio 3:1?", "What is the advantage of a cascade?", "Is widow's peak dominant or recessive?", "Why are X-linked disorders more common in males?"	A22, E24, B14, D18
Numbers to remember	46/23 chromosomes, $2^{23} \approx 8.4$ million, 10 bp and 3.4 Å per turn/bp (B-DNA), 7 transmembrane helices, ~60% of drugs, 10^8 amplification, 1200 → 180 finches.	C19, F5, E19

IN THE EXAM HALL

- Attempt every MCQ — unless your paper says there is negative marking.
- For genetics problems, write the genotypes of the parents and their gametes before anything else.
- Draw a 2 × 2 Punnett square per gene; multiply across genes.
- Hardy–Weinberg: always start from q^2 (the recessive phenotype) — never from the dominant phenotype.

CLASSIC TRAPS

- ‘Individuals evolve’ — false; populations evolve.
- Prophase vs prometaphase: the nuclear envelope breaks up in **prometaphase**.
- G protein ON = GTP, OFF = GDP.
- Adenylyl cyclase makes cAMP; phosphodiesterase destroys it.
- Sister chromatids separate in anaphase **II** (and in mitosis), homologues in anaphase **I**.

2 · Model Question Paper

Maximum marks: 60 · Time: 90 minutes · Answer all questions

Part I — Multiple choice (single correct) (1 mark each)

M1 MCQ [1 mark]

Mendel published his paper on inheritance in pea plants in:

(a) 1859 (b) 1866
(c) 1928 (d) 1953

M2 MCQ [1 mark]

Which of the following is a testcross?

(a) $Pp \times Pp$ (b) $PP \times PP$
(c) $Pp \times PP$ (d) $P_ \times pp$

M3 MCQ [1 mark]

Which of these was NOT one of the seven pea characters Mendel studied?

(a) Leaf shape (b) Pod colour
(c) Stem length (d) Seed shape

M4 MCQ [1 mark]

How many chromatids does a human cell contain in G_2?

(a) 23 (b) 46
(c) 92 (d) 184

M5 MCQ [1 mark]

In animal cells, the two centrosomes move apart to opposite ends of the cell during:

(a) telophase (b) G_1
(c) anaphase (d) prophase and prometaphase

M6 MCQ [1 mark]

Crossing over takes place between:

(a) sister chromatids of one chromosome
(b) non-sister chromatids of homologous chromosomes
(c) DNA and RNA
(d) two different non-homologous chromosomes

M7 MCQ [1 mark]

The karyotype of a person with Klinefelter syndrome is:

(a) 47, XXY (b) 45, X
(c) 47, +21 (d) 69, XXY ($3n$)

M8 MCQ [1 mark]

Two parents with normal blood clotting have a son with haemophilia. Most likely:

(a) the son received the allele on his Y chromosome
(b) the mother is a carrier
(c) both parents are affected
(d) the father is a carrier

M9 MCQ [1 mark]

In the adrenaline pathway, the second messenger is:

(a) the GPCR
(b) adrenaline
(c) glycogen
(d) cyclic AMP

M10 MCQ [1 mark]

Earl W. Sutherland's 1971 Nobel Prize was for work on:

(a) G proteins
(b) protein-structure prediction
(c) how hormones act, including the second messenger cAMP
(d) the double helix

M11 MCQ [1 mark]

The complete set of RNA transcripts in a cell is its:

(a) transcriptome
(b) genome
(c) metabolome
(d) proteome

M12 MCQ [1 mark]

Z-DNA has a negative rotation per base pair (−30°) because it is:

(a) left-handed
(b) made of RNA
(c) single-stranded
(d) denatured

M13 MCQ [1 mark]

The offspring of a female donkey and a male horse is a:

(a) mule
(b) liger
(c) hinny
(d) zebroid

M14 MCQ [1 mark]

Darwin's Observation #2 is that:

(a) species are fixed
(b) members of a population vary in inherited traits
(c) individuals evolve during their lifetime
(d) all species can produce more offspring than the environment can support

M15 MCQ [1 mark]

In a population at Hardy–Weinberg equilibrium, $q = 0.2$. The frequency of heterozygotes is:

(a) 0.04
(b) 0.16
(c) 0.32
(d) 0.64

Part II — One or more options correct (2 marks each)

Full marks only if all correct options, and no incorrect ones, are chosen.

M16 ONE OR MORE CORRECT [2 marks]

Which statements describe anaphase of mitosis?

(a) Nuclear envelopes re-form around each set of chromosomes
(b) Sister chromatids separate
(c) The cell elongates as non-kinetochore microtubules lengthen
(d) Kinetochore microtubules shorten

M17 ONE OR MORE CORRECT [2 marks]

Which are features of GPCR signalling?

(a) The receptor enters the nucleus to switch on genes
(b) Once switched on, the G protein stays active permanently
(c) The G protein binds GTP when it is activated
(d) Receptor has seven transmembrane domains

M18 ONE OR MORE CORRECT [2 marks]

Which of these would disturb Hardy–Weinberg equilibrium in a population?

(a) A new mutation
(b) Natural selection against one genotype
(c) The population shrinking to a few individuals
(d) Immigration of individuals from another population

M19 ONE OR MORE CORRECT [2 marks]

Which disorders are X-linked recessive?

(a) Albinism
(b) Down syndrome
(c) Duchenne muscular dystrophy
(d) Haemophilia

M20 ONE OR MORE CORRECT [2 marks]

According to the Big Data lecture, which statements are correct?

(a) Bioinformatics deals only with DNA sequences
(b) Bioinformatics began with the AlphaFold Nobel Prize in 2024
(c) Generating biological data is still too expensive to do on a large scale
(d) -Omics studies are complementary to one-protein-at-a-time studies

Part III — Match the following (4 marks each)

M21 MATCH THE FOLLOWING [4 marks]

Match the scientist with the discovery.

Column I	Column II
A. Thomas Hunt Morgan	1. Early protein-sequence database (PIR); pioneer of bioinformatics
B. Earl W. Sutherland	2. Evolution by natural selection (1859)
C. Margaret Dayhoff	3. X-linked white-eye gene in fruit flies
D. Charles Darwin	4. Transforming principle (1928)
	5. Mechanism of hormone action; cAMP (Nobel 1971)

M22 MATCH THE FOLLOWING [4 marks]

Match the chromosome-number term with its meaning.

Column I	Column II
A. Nondisjunction	1. More than two complete sets of chromosomes
B. Aneuploidy	2. Only one copy of a particular chromosome ($2n - 1$)
C. Polyploidy	3. Failure of chromosomes or chromatids to separate in meiosis
D. Monosomy	4. An abnormal number of a particular chromosome

M23 MATCH THE FOLLOWING [4 marks]

Match the population-genetics term with its meaning.

Column I	Column II
A. Gene flow	1. Contribution of an individual to the next generation's gene pool
B. Bottleneck effect	2. A few individuals colonise a new area
C. Founder effect	3. Transfer of alleles by migration of fertile individuals or gametes
D. Fitness	4. Drastic reduction in the size of a population

Part IV — Fill in the blank / True or False (1 mark each)

M24 FILL IN THE BLANKS [1 mark]

A cell spends about 90% of its cycle in _____.

M25 FILL IN THE BLANKS [1 mark]

Animal cells divide their cytoplasm with a cleavage _____; plant cells build a cell _____.

M26 TRUE / FALSE [1 mark]

Gametes are diploid.

Write True or False. If false, correct the statement.

M27 TRUE / FALSE [1 mark]

The Y chromosome is smaller than the X chromosome.

Write True or False. If false, correct the statement.

M28 FILL IN THE BLANKS [1 mark]

A single recessive allele on the X is expressed in a male because males are _____ for X-linked genes.

M29 TRUE / FALSE [1 mark]

Phosphodiesterase converts ATP into cAMP.

Write True or False. If false, correct the statement.

M30 FILL IN THE BLANKS [1 mark]

The suffix ‘-omics’ means _____.

M31 TRUE / FALSE [1 mark]

Populations, not individuals, evolve.

Write True or False. If false, correct the statement.

Part V — Numerical problems (3 marks each)

M32 NUMERICAL [3 marks]

For the cross *AaBbCc* × *aaBbcc* (independent genes, complete dominance), find (a) P(*aabbcc*), (b) P(offspring shows all three dominant traits).

M33 NUMERICAL [3 marks]

A woman with normal colour vision whose father was colour-blind marries a colour-blind man. Find (a) P(a daughter is colour-blind), (b) P(a son is colour-blind), (c) P(any child is colour-blind).

M34 NUMERICAL [3 marks]

In a population of 1000 at Hardy–Weinberg equilibrium, 160 individuals show the recessive phenotype. How many individuals are expected to be heterozygous, and how many homozygous dominant?

Part VI — Short answer (3 marks each)

M35 SHORT ANSWER [3 marks]

Trace how ONE adrenaline molecule binding to a liver cell leads to the release of about 10^8 molecules of glucose 1-phosphate. Name the three stages of signalling.

M36 SHORT ANSWER [3 marks]

Describe what happened to the medium ground finches of Daphne Major during the 1977 drought, and explain what it shows about who evolves.

3 · Section A — Heredity and Mendel's laws

Lectures 15–16 · Campbell Ch. 14.1 · 23 questions

A1 MCQ

Which of the following was **NOT** a reason Mendel chose the garden pea as his model system?

(a) Its characters vary continuously, which makes blending easy to measure
(b) It exists in many varieties with distinct heritable features
(c) It has a short generation time and gives many offspring per mating
(d) Its pollen-producing and egg-bearing organs are well separated, so crosses are easy to control

A2 MCQ

In Mendel's terminology, 'flower colour' is a _____ and 'purple' is a _____.

(a) gene; allele
(b) phenotype; genotype
(c) trait; character
(d) character; trait

A3 MCQ

Mendel crossed true-breeding purple-flowered plants with true-breeding white-flowered plants. The F_1 plants were:

(a) all purple
(b) all pink
(c) 1 purple : 1 white
(d) 3 purple : 1 white

A4 MCQ

Mendel counted 705 purple-flowered and 224 white-flowered F_2 plants. This ratio is closest to:

(a) 3 : 1
(b) 1 : 1
(c) 9 : 3 : 3 : 1
(d) 2 : 1

A5 MCQ

The reappearance of white flowers in the F_2 generation was strong evidence **against**:

(a) the existence of dominant alleles
(b) the particulate hypothesis
(c) the law of segregation
(d) the blending hypothesis

A6 MCQ

According to the law of segregation, a heterozygous *Pp* plant produces gametes in the proportion:

(a) all *P*, because *P* is dominant
(b) ½ *P* : ½ *p*
(c) ¾ *P* : ¼ *p*
(d) all *Pp*

A7 MCQ

The physical basis of Mendel's law of segregation is:

(a) fusion of gametes at fertilization
(b) separation of sister chromatids during mitotic anaphase
(c) replication of DNA during S phase
(d) separation of homologous chromosomes during anaphase I of meiosis

A8 MCQ

Independent assortment is expected for two genes that are:

(a) only in organisms that reproduce asexually

(b) close together on the same chromosome

(c) on different chromosomes, or far apart on the same chromosome

(d) both on the Y chromosome

A9 MATCH THE FOLLOWING

Match each scientist (Column I) with the contribution (Column II).

Column I	Column II
A. Mendel (1866)	1. DNA identified as the transforming substance
B. Griffith (1928)	2. Double-helix structure of DNA
C. Avery, MacLeod & McCarty (1944)	3. Rules of inheritance from pea crosses
D. Watson & Crick (1953)	4. ‘Transforming principle’ in bacteria
	5. Discovery of G proteins

A10 MATCH THE FOLLOWING

Match the breeding term with its meaning.

Column I	Column II
A. P generation	1. Offspring of F_1 plants that self- or cross-pollinate
B. F_1 generation	2. Crossing two contrasting true-breeding varieties
C. F_2 generation	3. Observable property of an organism
D. Hybridization	4. True-breeding parents
E. Phenotype	5. Hybrid offspring of the P cross

A11 FILL IN THE BLANKS

Mendel’s heritable ‘factor’ (or ‘unit’) is today called a _____; its alternative versions are called _____.

A12 FILL IN THE BLANKS

In the F_2 of a monohybrid cross the phenotypic ratio is _____ and the genotypic ratio is _____.

A13 FILL IN THE BLANKS

An organism’s full hereditary information is its _____, which is made of _____. Traits may be qualitative, such as _____, or quantitative, such as _____.

A14 TRUE / FALSE

In the F_1 hybrids, the white-flower factor is diluted and permanently lost.

Write True or False. If false, correct the statement.

A15 TRUE / FALSE

PP and *Pp* pea plants have the same phenotype but different genotypes.

Write True or False. If false, correct the statement.

A16 TRUE / FALSE

The law of segregation describes how alleles of two different genes are inherited relative to each other.

Write True or False. If false, correct the statement.

A17 ASSERTION – REASON

Assertion (A): All F_1 plants from a purple × white cross have purple flowers.
Reason (R): The purple allele is dominant, so a single copy of it is enough to produce purple colour.

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

A18 ASSERTION – REASON

Assertion (A): Two siblings (other than identical twins) are never genetically identical.
Reason (R): Every child receives half of its chromosomes from each parent.

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

A19 NUMERICAL

A cross *Pp* × *Pp* gives 800 F_2 plants. (a) How many are expected to be white? (b) What fraction of the **purple** plants are expected to be heterozygous?

A20 NUMERICAL

For the dihybrid cross *YyRr* × *YyRr* (Y = yellow, R = round, genes on different chromosomes): (a) what fraction of offspring is yellow and round? (b) green and wrinkled? (c) how many different kinds of gametes does a *YyRr* plant make?

A21 NUMERICAL

How many genetically different gametes can an *AaBbCcDd* individual produce if all four genes are on different chromosomes?

A22 SHORT ANSWER

“Why is the ratio 3 : 1? What does this mean?” Answer in 3–4 lines.

A23 SHORT ANSWER

Why did Mendel follow each character to at least the F_2 generation instead of stopping at F_1?

Section B — Mendel and the gene idea: probability, dominance, pedigrees

Lecture 17 · Campbell Ch. 14.1–14.4 · 23 questions

B1 MCQ

A Punnett square is used to:

(a) show all possible combinations of sperm and egg, to predict the results of a cross
(b) draw a family tree across generations
(c) locate a gene on a chromosome
(d) measure how often crossing over occurs

B2 MCQ

Which genotype is heterozygous?

(a) Jj
(b) HH
(c) TT
(d) zz

B3 MCQ

At the molecular level, why does a Pp pea plant have purple flowers?

(a) The P protein destroys the protein made by p
(b) The two alleles blend, and purple is the darker colour
(c) One functional P allele makes enough active enzyme to synthesise the purple pigment
(d) The p allele is deleted from heterozygous cells

B4 MCQ

The specific position of a gene on a chromosome is called its:

(a) allele
(b) centromere
(c) chromatid
(d) locus

B5 MCQ

Red snapdragons (C^RC^R) × white snapdragons (C^WC^W) give all pink F_1. This is an example of:

(a) pleiotropy
(b) incomplete dominance
(c) complete dominance
(d) codominance

B6 MCQ

People with blood group AB carry both A and B carbohydrates on their red blood cells. The relationship between I^A and I^B is:

(a) epistasis
(b) complete dominance of I^A
(c) incomplete dominance
(d) codominance

B7 MCQ

With the three ABO alleles I^A, I^B and i, the number of possible genotypes and blood-group phenotypes is:

(a) 3 genotypes, 3 phenotypes
(b) 4 genotypes, 4 phenotypes
(c) 6 genotypes, 4 phenotypes
(d) 9 genotypes, 4 phenotypes

B8 MCQ

Pleiotropy means:

(a) many genes together control one character
(b) a gene has more than two alleles
(c) one gene has multiple phenotypic effects
(d) the heterozygote looks intermediate

B9 MCQ

From *Rr* × *Rr*, the probability of a heterozygous offspring is worked out as 1/4 + 1/4 = 1/2. Which rule is being used?

(a) addition rule
(b) Hardy–Weinberg principle
(c) multiplication rule
(d) law of independent assortment

B10 MCQ

Why are humans poor subjects for classical breeding experiments?

(a) Generation time is too long
(b) Couples produce relatively few offspring
(c) Controlled breeding experiments are unacceptable
(d) All of the above

B11 ONE OR MORE CORRECT

Which statements about dominance are correct?

(a) In codominance, both alleles affect the phenotype in separate, distinguishable ways
(b) A dominant allele does not subdue or inhibit a recessive allele; alleles don't interact that way
(c) Dominant alleles are always more common in a population than recessive alleles
(d) Whether alleles look completely dominant, incompletely dominant or codominant can depend on the level at which the phenotype is examined

B12 MATCH THE FOLLOWING

Match the inheritance pattern with the example.

Column I	Column II
A. Complete dominance	1. Pink F_1 snapdragons from red × white
B. Incomplete dominance	2. ABO gene having three alleles in the population
C. Codominance	3. Purple vs white pea flowers
D. Multiple alleles	4. One gene (e.g. the sickle-cell allele) affecting many characters
E. Pleiotropy	5. Both A and B carbohydrates on the red cells of I^AI^B people

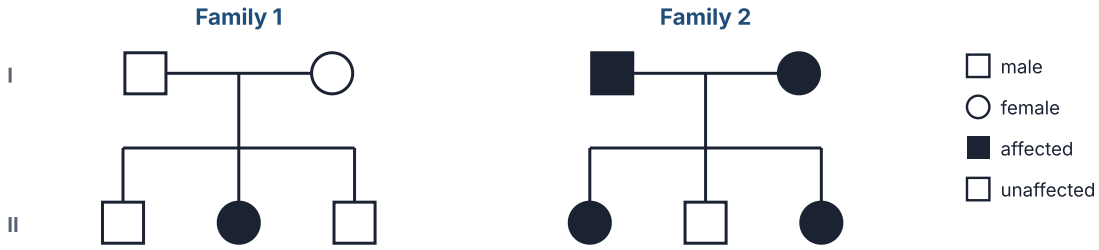
B13 MATCH THE FOLLOWING

Match the genotype with the blood group.

Column I	Column II
A. I^Ai	1. O
B. I^BI^B	2. AB
C. I^AI^B	3. A
D. ii	4. B

B14 SHORT ANSWER

Study the two pedigrees. (a) In Family 1, is the trait dominant or recessive? (b) In Family 2? (c) Give the genotypes of the Family 1 parents (use *A/a*). (d) What is the probability that the next child of the Family 1 parents is affected?



B15 NUMERICAL

For the cross *PpYyRr* × *Ppyyrr*, what fraction of the offspring is predicted to show the recessive phenotype for **at least two** of the three characters? (This is the problem on the lecture slides.)

B16 NUMERICAL

For *AaBbCc* × *AaBbCc* (independent genes), find the probability of an offspring that is (a) *AABBCC*, (b) *AaBbCc*, (c) *aabbcc*.

B17 NUMERICAL

Two pink snapdragons (C^RC^W) are crossed and produce 240 seedlings. How many red, pink and white plants are expected?

B18 NUMERICAL

A man with genotype I^Ai and a woman with genotype I^Bi have children. Find (a) P(child is AB), (b) P(child is O), (c) P(their first two children are both O).

B19 MCQ

A mother has blood group O and her child has blood group A. The father's blood group could be:

(a) B or O
(b) A or AB
(c) O only
(d) B only

B20 NUMERICAL

Albinism is recessive. Two normally pigmented parents are both carriers (*Aa*). Find (a) P(albino child), (b) P(a normally pigmented child is a carrier), (c) P(their first two children are both normal).

B21 SHORT ANSWER

Using the pea flower-colour example, explain why "one functional allele suffices".

B22 TRUE / FALSE

Pedigrees can be used to predict the probability of specific phenotypes in a couple's future children.

Write True or False. If false, correct the statement.

B23 FILL IN THE BLANKS

Multiplication rule: the probability that two or more _____ events occur together equals the _____ of their individual probabilities.

Section C — Cell cycle, mitosis and meiosis

Lecture 18 · Campbell Ch. 12 and 13 · 26 questions

C1 MCQ

Human somatic cells contain _____ chromosomes, and human gametes contain _____.

(a) 23; 46
(b) 46; 46
(c) 46; 23
(d) 92; 46

C2 MCQ

During which phase of the cell cycle are the chromosomes duplicated?

(a) G_1
(b) S
(c) G_2
(d) M

C3 MCQ

In which stage of mitosis does the nuclear envelope fragment?

(a) Prophase
(b) Prometaphase
(c) Metaphase
(d) Telophase

C4 MCQ

Kinetochores are:

(a) protein complexes associated with the centromere of each chromatid
(b) vesicles that build the cell plate
(c) radial arrays of short microtubules around each centrosome
(d) the regions where crossing over happens

C5 MCQ

The metaphase plate is:

(a) the nuclear envelope at metaphase
(b) an imaginary plane midway between the two spindle poles
(c) a membrane that divides the cell in two
(d) the cell plate of a dividing plant cell

C6 MCQ

During anaphase, chromosomes move to the poles mainly because kinetochore microtubules:

(a) push against the plasma membrane
(b) fuse with the nuclear envelope
(c) lengthen by adding tubulin at their kinetochore ends
(d) shorten by depolymerizing at their kinetochore ends

C7 MCQ

The mitotic spindle consists of:

(a) chromosomes and kinetochores only
(b) actin microfilaments and myosin
(c) the nuclear envelope and nucleoli
(d) the centrosomes, the spindle microtubules and the asters

C8 MCQ

Synapsis of homologous chromosomes and crossing over occur during:

(a) prophase of mitosis
(b) metaphase II of meiosis
(c) prophase I of meiosis
(d) anaphase I of meiosis

C9 MCQ

A cell with $2n = 6$ completes meiosis I. Each daughter cell has:

(a) 3 chromosomes, each of two sister chromatids
(b) 6 chromosomes, each of one chromatid
(c) 6 chromosomes, each of two sister chromatids
(d) 3 chromosomes, each of one chromatid

C10 MCQ

Which of these is **NOT** a source of genetic variation in sexually reproducing organisms?

(a) Crossing over
(b) Separation of sister chromatids in mitosis
(c) Independent assortment of homologues
(d) Random fertilization

C11 MCQ

Cytokinesis in a plant cell takes place by:

(a) a cell plate formed from vesicles that fuse in the middle of the cell
(b) pinching of the nuclear envelope
(c) depolymerization of spindle microtubules
(d) a cleavage furrow formed by a ring of actin and myosin

C12 ONE OR MORE CORRECT

Which events occur in meiosis but **NOT** in mitosis?

(a) Formation of a spindle from microtubules
(b) Separation of sister chromatids
(c) Crossing over between non-sister chromatids
(d) Synapsis of homologous chromosomes

C13 MATCH THE FOLLOWING

Match each stage of mitosis with its key event.

Column I	Column II
A. Prophase	1. Sister chromatids separate and move to opposite poles
B. Prometaphase	2. Nuclear envelope fragments; microtubules attach to kinetochores
C. Metaphase	3. Nuclear envelopes and nucleoli re-form; chromosomes decondense
D. Anaphase	4. Chromatin condenses, nucleoli disappear, spindle begins to form
E. Telophase	5. Chromosomes line up at the metaphase plate

C14 MATCH THE FOLLOWING

Match the phase of the cell cycle with what happens in it.

Column I	Column II
A. G_1	1. Cytoplasm divides, producing two daughter cells
B. S	2. Chromosomes duplicate
C. G_2	3. First part of interphase (growth)
D. M (mitosis)	4. Last part of interphase, just before division
E. Cytokinesis	5. Chromosomes are distributed to two daughter nuclei

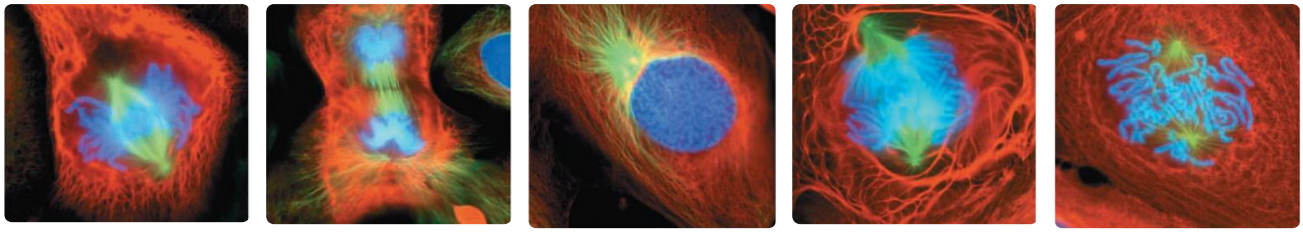
C15 ARRANGE IN ORDER

Arrange the stages of mitosis in the correct order.

(i) Anaphase
(ii) Prometaphase
(iii) Telophase
(iv) Prophase
(v) Metaphase

C16 SHORT ANSWER

The five micrographs A–E (one per stage of mitosis) are shown in jumbled order. Name the stage shown in each.



A B C D E

Fluorescence micrographs from the lecture slides (Campbell Fig. 12.7): chromosomes blue, microtubules green, actin red.

C17 NUMERICAL

A human cell in G_1 has a DNA content of x. Give the chromosome number and DNA content (in terms of x) of: (a) the same cell in G_2; (b) each daughter cell after mitosis; (c) each cell after meiosis I; (d) each gamete after meiosis II.

C18 NUMERICAL

A species has $2n = 8$. (a) How many chromosome combinations are possible in its gametes from independent assortment alone? (b) How many tetrads (bivalents) form in prophase I? (c) How many chromatids are present at metaphase of mitosis?

C19 NUMERICAL

In humans ($n = 23$), how many chromosome combinations can one person's gametes have from independent assortment alone? Roughly how many zygote combinations can one couple produce (ignoring crossing over)?

C20 TRUE / FALSE

DNA is replicated again between meiosis I and meiosis II.

Write True or False. If false, correct the statement.

C21 TRUE / FALSE

Once sister chromatids separate, each one is called a chromosome.

Write True or False. If false, correct the statement.

C22 TRUE / FALSE

Mitosis is the division of the cytoplasm of a cell.

Write True or False. If false, correct the statement.

C23 FILL IN THE BLANKS

The narrow ‘waist’ of a duplicated chromosome, where the sister chromatids are most closely attached, is the _____. Chromatin is a complex of _____ and _____.

C24 ASSERTION – REASON

Assertion (A): Meiosis is called a reduction division.

Reason (R): DNA is replicated a second time between meiosis I and meiosis II.

(a) Both A and R are true, and R is the correct explanation of A

(b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false

(d) A is false, but R is true

C25 SHORT ANSWER

Give four differences between mitosis and meiosis.

C26 SHORT ANSWER

Distinguish homologous chromosomes from sister chromatids.

Section D — Chromosomal basis of inheritance

Lecture 19 · Campbell Ch. 15 · 23 questions

D1 MCQ

The chromosome theory of inheritance states that:

(a) genes are made of protein and are found in the cytoplasm
(b) chromosomes blend in the offspring
(c) each chromosome carries only one gene
(d) Mendelian genes have specific loci on chromosomes, and it is the chromosomes that undergo segregation and independent assortment

D2 MCQ

Morgan crossed a red-eyed (wild-type) female fruit fly with a white-eyed male. The F_2 showed 3 red : 1 white, but all white-eyed flies were male. He concluded that:

(a) the white-eye allele is dominant
(b) the eye-colour gene is on the X chromosome, with no corresponding locus on the Y
(c) the eye-colour gene is on the Y chromosome
(d) eye colour shows incomplete dominance

D3 MCQ

In *Drosophila* notation, w^+ represents:

(a) a Y-linked allele
(b) the mutant white-eye allele
(c) the wild-type (red-eye) allele
(d) a dominant lethal allele

D4 MCQ

For X-linked genes, a human male is described as:

(a) homozygous
(b) hemizygous
(c) heterozygous
(d) polyploid

D5 MCQ

Which of these is an X-linked recessive disorder?

(a) Haemophilia
(b) Albinism
(c) Turner syndrome
(d) Down syndrome

D6 MCQ

If homologous chromosomes fail to separate in meiosis I (nondisjunction), the four gametes will have:

(a) $n+1$, $n-1$, n, n
(b) $2n$, $2n$, 0, 0
(c) n, n, n, n
(d) $n+1$, $n+1$, $n-1$, $n-1$

D7 MCQ

If sister chromatids fail to separate in one of the two cells in meiosis II, the gametes will have:

(a) all $n+1$
(b) all n
(c) $n+1$, $n-1$, n, n
(d) $n+1$, $n+1$, $n-1$, $n-1$

D8 MCQ

Which statement about Turner syndrome is correct?

(a) Trisomy 21
(b) Karyotype $3n$
(c) Karyotype XXY; male
(d) Karyotype X0; female; the only known viable monosomy in humans

D9 MCQ

A zygote with one extra copy of a particular chromosome ($2n + 1$) is called:

(a) monosomic
(b) trisomic
(c) triploid
(d) tetraploid

D10 MCQ

The SRY gene is:

(a) on the Y chromosome; its protein triggers development of male features (testes)
(b) a gene for red–green colour vision
(c) on the X chromosome; it determines female development
(d) on chromosome 21

D11 MCQ

Which statement about human sex chromosomes is correct?

(a) Females are XY and males are XX
(b) An ovum may carry an X or a Y chromosome
(c) Only the ends of the Y chromosome have regions homologous with the X
(d) The Y chromosome carries genes for many characters unrelated to sex

D12 MATCH THE FOLLOWING

Match the condition with its chromosome make-up.

Column I	Column II
A. Down syndrome	1. XXY (male)
B. Klinefelter syndrome	2. X0 (female)
C. Turner syndrome	3. Trisomy 21
D. Triploidy	4. $2n - 1$
E. Monosomy	5. $3n$

D13 NUMERICAL

A colour-blind man (X^nY) marries a woman who is homozygous for normal vision (X^NX^N). Find (a) P(a daughter is colour-blind), (b) the genotype of every daughter, (c) P(a son is colour-blind).

D14 NUMERICAL

A carrier woman (X^NX^n) marries a man with normal vision (X^NY). Find (a) P(a son is colour-blind), (b) P(a daughter is a carrier), (c) P(their next child — sex not yet known — is colour-blind).

D15 NUMERICAL

A carrier woman (X^NX^n) marries a colour-blind man (X^nY). (a) What is the chance that any one child is colour-blind? (b) What are the normal-vision daughters? (c) What are the normal-vision sons?

D16 NUMERICAL

Morgan's reciprocal cross: a white-eyed female (X^wX^w) is mated with a red-eyed male ($X^{w+}Y$). Predict the F_1 phenotypes of each sex.

D17 NUMERICAL

In Morgan's original F_2 (from F_1 red female × F_1 red male), out of 400 flies, how many white-eyed flies are expected, and how many of them are female?

D18 ASSERTION – REASON

Assertion (A): X-linked recessive disorders are much more common in males than in females.

Reason (R): Males are hemizygous, so a single recessive allele on their X chromosome is expressed.

(a) Both A and R are true, and R is the correct explanation of A

(b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false

(d) A is false, but R is true

D19 ASSERTION – REASON

Assertion (A): Polyploid organisms are usually more normal in appearance than aneuploid ones.

Reason (R): Polyploidy is common in plants.

(a) Both A and R are true, and R is the correct explanation of A

(b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false

(d) A is false, but R is true

D20 TRUE / FALSE

Polyploidy is common in animals but rare in plants.

Write True or False. If false, correct the statement.

D21 TRUE / FALSE

The frequency of Down syndrome increases with the age of the mother.

Write True or False. If false, correct the statement.

D22 FILL IN THE BLANKS

In mammals, females are _____ and males are _____. Every ovum carries an _____ chromosome, while _____ % of sperm carry a Y.

D23 SHORT ANSWER

Explain why a colour-blind father can never pass colour blindness to his son, yet every one of his daughters is a carrier.

Section E — Cell communication and signal transduction

Lecture 20 · Campbell Ch. 11 · 25 questions

E1 MCQ

When food is scarce, cells of the soil bacterium *Myxococcus xanthus*:

(a) switch mating type
(b) fuse to form a diploid cell
(c) divide faster
(d) send out a chemical signal, aggregate and form fruiting bodies containing spores

E2 MCQ

Quorum sensing allows bacteria to:

(a) sense the local population density through the concentration of signalling molecules
(b) move towards food
(c) exchange DNA by conjugation
(d) detect light

E3 MCQ

In baker's yeast, the 'a factor' secreted by type-a cells binds to receptors on:

(a) the a cell's own nucleus
(b) type-α cells
(c) bacterial cells
(d) other type-a cells

E4 MCQ

Fusion of an a cell and an α cell produces:

(a) two haploid a cells
(b) a spore-forming fruiting body
(c) an a/α diploid cell containing all the genes of both cells
(d) a gamete

E5 MCQ

Junctions that directly connect the cytoplasm of adjacent cells are called _____ in animals and _____ in plants.

(a) gap junctions; plasmodesmata
(b) tight junctions; stomata
(c) synapses; cell walls
(d) plasmodesmata; gap junctions

E6 MCQ

A growth factor released by one cell that stimulates nearby target cells to grow and divide is an example of:

(a) cell–cell recognition
(b) endocrine signalling
(c) paracrine signalling
(d) synaptic signalling

E7 MCQ

Hormones that travel through the bloodstream and act only on cells carrying the right receptors illustrate:

(a) quorum sensing
(b) paracrine signalling
(c) gap-junction signalling
(d) long-distance (endocrine) signalling

E8 MCQ

The correct order of the three stages of cell signalling is:

(a) transduction → reception → response
(b) reception → transduction → response
(c) response → reception → transduction
(d) reception → response → transduction

E9 MCQ

A G protein is in its active (ON) state when it is bound to:

(a) ATP
(b) cAMP
(c) GTP
(d) GDP

E10 MCQ

A G protein switches itself off by:

(a) binding a second signalling molecule
(b) being degraded in the lysosome
(c) leaving the plasma membrane
(d) hydrolysing its bound GTP to GDP and P_i (GTPase activity)

E11 MCQ

G protein-coupled receptors share a common architecture of:

(a) seven transmembrane domains
(b) a single ion channel pore
(c) an intracellular DNA-binding domain
(d) two subunits that dimerize

E12 MCQ

Adenylyl cyclase converts:

(a) glycogen → glucose 1-phosphate
(b) ATP → cAMP
(c) GTP → GDP
(d) cAMP → AMP

E13 MCQ

Which pair received the 1994 Nobel Prize for discovering G proteins?

(a) Alfred Gilman and Martin Rodbell
(b) Avery and McCarty
(c) Watson and Crick
(d) Earl Sutherland and Thomas Hunt Morgan

E14 MCQ

Approximately what fraction of all medicines in use today act on GPCRs?

(a) ~6%
(b) ~20%
(c) ~60%
(d) ~95%

E15 ONE OR MORE CORRECT

Which of the following can be a cellular **response** at the end of a signalling pathway?

(a) A change in the cell's DNA sequence
(b) Rearrangement of the cytoskeleton
(c) Activation of specific genes (gene expression)
(d) Catalysis of a reaction by an activated enzyme

E16 MATCH THE FOLLOWING

Match the mode of signalling with its description.

Column I	Column II
A. Gap junctions	1. Hormone carried by blood to distant targets
B. Plasmodesmata	2. Channels joining the cytoplasm of adjacent plant cells
C. Paracrine	3. Local regulator (e.g. growth factor) acts on nearby cells
D. Synaptic	4. Neurotransmitter diffuses across a synapse
E. Endocrine	5. Channels joining the cytoplasm of adjacent animal cells

E17 MATCH THE FOLLOWING

Match the molecule with its role in the adrenaline (epinephrine) pathway.

Column I	Column II
A. Adrenaline	1. Second messenger
B. cAMP	2. Converts cAMP → AMP
C. Adenylyl cyclase	3. First messenger (hormone)
D. Phosphodiesterase	4. Activated by cAMP; phosphorylates other proteins
E. Protein kinase A	5. Converts ATP → cAMP

E18 ARRANGE IN ORDER

Arrange the events of GPCR signalling in order.

(i) The activated receptor binds an inactive G protein
(ii) A signalling molecule (ligand) binds the GPCR on the outside
(iii) GTPase activity hydrolyses GTP; the G protein becomes inactive again
(iv) The receptor changes shape (conformation)
(v) The G protein swaps GDP for GTP, leaves the receptor and activates an enzyme

E19 NUMERICAL

In the adrenaline cascade on the slide, 1 adrenaline molecule → 10^2 active G proteins → 10^2 adenylyl cyclase → 10^4 cAMP → 10^4 PKA → 10^5 phosphorylase kinase → 10^6 glycogen phosphorylase → 10^8 glucose 1-phosphate. (a) What is the overall amplification? (b) Name the three steps that each amplify the signal 100-fold. (c) How many glucose 1-phosphate molecules result per active PKA molecule?

E20 ASSERTION – REASON

Assertion (A): In cholera, intestinal cells keep producing cAMP and secrete large amounts of salt and water.

Reason (R): Cholera toxin modifies a G protein so that it cannot hydrolyse GTP and stays active.

(a) Both A and R are true, and R is the correct explanation of A

(b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false

(d) A is false, but R is true

E21 ASSERTION – REASON

Assertion (A): One adrenaline molecule bound to a liver cell can release only a few molecules of glucose 1-phosphate.

Reason (R): In a signalling cascade, each activated molecule activates many molecules of the next step.

(a) Both A and R are true, and R is the correct explanation of A

(b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false

(d) A is false, but R is true

E22 TRUE / FALSE

Signalling between cells is found only in multicellular organisms.

Write True or False. If false, correct the statement.

E23 FILL IN THE BLANKS

To ‘transduce’ means to _____. In the doorbell analogy, the receptor acts as the _____, the relay molecules as the _____, and the cascade as the _____.

E24 SHORT ANSWER

“What is the advantage of a cascade?” (slide A → B → E → F). Give three advantages.

E25 SHORT ANSWER

A drug blocks phosphodiesterase in liver cells responding to adrenaline. Predict the effect, with reasoning.

Section F — Big data in biology and bioinformatics

Lecture 21 · Lecture slides; Campbell 16.1, 21.2 · 24 questions

F1 MCQ

The suffixes ‘-ome’ and ‘-omics’ mean, respectively:

(a) a single gene; many genes
(b) the study of; the complete set
(c) the complete set; the study of an -ome
(d) protein; RNA

F2 MCQ

The proteome is:

(a) the set of all metabolites
(b) the complete set of RNA transcripts
(c) the community of microbes in the gut
(d) the complete set of proteins of a cell or organism

F3 MCQ

Computer-aided analysis of protein data — the beginning of bioinformatics, around 1960 — was pioneered by:

(a) Watson, Crick and Franklin
(b) Margaret Dayhoff, Richard Eck and Robert Ledley
(c) Gilman and Rodbell
(d) Hassabis and Jumper

F4 MCQ

The metric prefix ‘exa-’ stands for:

(a) 10^{12}
(b) 10^{15}
(c) 10^{18}
(d) 10^{21}

F5 MCQ

For B-DNA (the common form), the table on the slide gives:

(a) 11 bp per turn, 2.55 Å between base pairs
(b) 12 bp per turn, left-handed
(c) 10 bp per turn, 3.4 Å between base pairs, diameter 20 Å
(d) 10 bp per turn, diameter 23 Å

F6 MCQ

Z-DNA differs from B-DNA because it is:

(a) made of RNA
(b) right-handed, with 11 bp per turn
(c) single-stranded
(d) left-handed, with about 12 bp per turn

F7 MCQ

Methods that derive a *post hoc* understanding of what a trained model has learnt are called:

(a) databases
(b) explainable methods
(c) black boxes
(d) interpretable models

F8 MCQ

Which of these is **NOT** a reason for comparing biological sequences?

(a) To weigh a protein directly
(b) To find whether genes or proteins are evolutionarily related
(c) To find similar regions such as catalytic or binding sites
(d) To infer the function of a sequence from a better-characterised one

F9 MCQ

'Resequencing' an individual's genome means:

(a) mapping the sequencing reads to a known reference genome to find differences such as SNPs and indels
(b) sequencing a new species for the first time
(c) translating DNA into protein
(d) repeating Sanger sequencing on the same sample

F10 MCQ

The 2024 Nobel Prize in Chemistry was awarded to:

(a) Gilman and Rodbell for G proteins
(b) Dayhoff for the PIR database
(c) Watson and Crick for the double helix
(d) David Baker, Demis Hassabis and John Jumper for computational protein design and structure prediction (AlphaFold)

F11 MCQ

From the 'methods for extracting structural information' chart, which technique **cannot** resolve individual atoms?

(a) Electron microscopy
(b) Light microscopy
(c) X-ray crystallography
(d) NMR

F12 ONE OR MORE CORRECT

According to the AlphaFold slide, which inputs feed the deep neural network that predicts a protein's 3D structure?

(a) Known 3D structures of proteins (whatever is available)
(b) The mass spectrum of the protein
(c) The amino-acid sequence of the target protein
(d) Its evolutionary history (related sequences)

F13 MATCH THE FOLLOWING

Match the '-ome' with what it is the complete set of.

Column I	Column II
A. Genome	1. All the proteins
B. Transcriptome	2. All the small-molecule metabolites
C. Proteome	3. All the DNA (hereditary information)
D. Metabolome	4. The microbial community (and its genes)
E. Microbiome	5. All the RNA transcripts

F14 MATCH THE FOLLOWING

Match the metric prefix with its power of ten.

Column I	Column II
A. kilo-	1. 10^{21}
B. giga-	2. 10^{9}
C. peta-	3. 10^{3}
D. exa-	4. 10^{18}
E. zetta-	5. 10^{15}

F15 MATCH THE FOLLOWING

Bioinformatics sits where three fields meet. Match each field with its role.

Column I	Column II
A. Statistics	1. Develop algorithms; storage and retrieval of data
B. Computer science	2. Interpretation, hypothesis testing, context for data generation
C. Biology (domain knowledge)	3. Choice of metrics, pattern extraction, validity of inferences

F16 NUMERICAL

Using the B-DNA rise of 3.4 Å per base pair, how long would the DNA of one haploid human genome (3.0×10^9 bp) be if stretched out? How much DNA is in one diploid cell?

F17 NUMERICAL

A 1000-bp stretch of DNA is in the B form. (a) How many helical turns does it make? (b) What is its length? (c) How many turns would it make in the A form?

F18 NUMERICAL

Each DNA base (A, C, G or T) can be stored in 2 bits. How much storage (in megabytes) does one 3.0×10^9-base human genome need? (1 byte = 8 bits.)

F19 TRUE / FALSE

'-Omics' studies make one-protein-at-a-time studies unnecessary.

Write True or False. If false, correct the statement.

F20 TRUE / FALSE

Data generation in biology has become affordable, which is one reason biology produces very large data sets.

Write True or False. If false, correct the statement.

F21 FILL IN THE BLANKS

SNP stands for _____; an indel is an _____ or _____; ChIP-Seq maps the binding sites of _____ on the genome.

F22 ASSERTION – REASON

Assertion (A): Genomics generates data on a scale comparable to astronomy and YouTube.
Reason (R): Bioinformatics began around 1960 with computer-aided analysis of protein sequences.

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

F23 SHORT ANSWER

Give the two definitions of bioinformatics by Higgs and Attwood.

F24 SHORT ANSWER

Why is the 'black box' nature of AI/ML a concern in biology, and what are the two approaches to it?

Section G — Evolution and natural selection

Lecture 22 · Campbell Ch. 22, 23 (opening), 24.1 · 21 questions

G1 MCQ

Darwin's *On the Origin of Species by Means of Natural Selection* was published in:

(a) 1809
(b) 1859
(c) 1866
(d) 1953

G2 MCQ

Darwin's three broad observations about nature were:

(a) reception, transduction and response
(b) mutation, drift and gene flow
(c) the unity of life, the diversity of life, and the match between organisms and their environments
(d) segregation, independent assortment and dominance

G3 MCQ

According to the biological species concept, a species is a group of populations whose members:

(a) can interbreed in nature and produce viable, fertile offspring
(b) look alike
(c) have identical DNA
(d) live in the same place

G4 MCQ

A male donkey × female horse gives a mule; a female donkey × male horse gives a hinny. Both are sterile. This shows that:

(a) donkeys evolved from horses recently
(b) horses and donkeys are the same species
(c) hybrids are always fertile
(d) horses and donkeys are separate species

G5 MCQ

The biological species concept cannot be applied to:

(a) mammals
(b) birds
(c) organisms that reproduce only asexually (divide to multiply)
(d) flowering plants

G6 MCQ

An adaptation is:

(a) any change an individual makes during its lifetime
(b) an inherited characteristic that enhances survival and reproduction in a specific environment
(c) a random mutation
(d) a learned behaviour

G7 MCQ

Malthus influenced Darwin with the idea that:

(a) populations increase exponentially while food supply increases only linearly
(b) species are fixed
(c) acquired characteristics are inherited
(d) the Earth is only a few thousand years old

G8 MCQ

After the 1977 drought on Daphne Major, the average beak depth of the medium ground finch population increased because:

(a) each bird's beak grew larger to crack hard seeds
(b) small-beaked birds migrated away
(c) the drought caused mutations for larger beaks
(d) birds with deeper beaks could crack the plentiful large, hard seeds, survived and reproduced

G9 MCQ

Which statement is correct?

(a) Natural selection creates the variation it acts on
(b) Natural selection acts on individuals, but it is populations that evolve over time
(c) Natural selection is goal-directed
(d) Individuals evolve during their lifetime

G10 MCQ

According to the lecture, the enzyme that replicates DNA makes roughly:

(a) 1 error per genome
(b) 1 error in every 10 nucleotides
(c) 1 error in every 10^4–10^5 nucleotides added
(d) no mistakes

G11 MCQ

Methicillin kills *Staphylococcus aureus* by:

(a) inhibiting a protein that the bacteria use to build their cell walls
(b) blocking ribosomes
(c) dissolving the cell membrane
(d) blocking DNA replication

G12 MATCH THE FOLLOWING

Match the Galápagos finch with its beak and diet.

Column I	Column II
A. Large ground finch (*Geospiza magnirostris*)	1. Narrow, pointed beak — grasping insects
B. Cactus ground finch (*Geospiza scandens*)	2. Large but short beak — cracking seeds that fall from trees
C. Green warbler finch (*Certhidea olivacea*)	3. Long, sharp beak — tearing cactus flowers and pulp

G13 MATCH THE FOLLOWING

Match each of Darwin's four components of natural selection with its description.

Column I	Column II
A. Variation	1. Individuals with traits suited to the struggle for local resources leave more offspring
B. Inheritance	2. More offspring each year than local resources can support
C. High rate of population growth	3. Individuals differ in size, colour, markings, voice…
D. Differential survival and reproduction	4. Some traits are passed consistently from parent to offspring

G14 ASSERTION – REASON

Assertion (A): After the drought, the proportion of large-beaked finches in the population increased.

Reason (R): Individual finches grew larger beaks during the drought.

(a) Both A and R are true, and R is the correct explanation of A

(b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false

(d) A is false, but R is true

G15 ASSERTION – REASON

Assertion (A): A trait favoured in one place or time may be useless or harmful in another.

Reason (R): Environmental factors vary from place to place and over time.

(a) Both A and R are true, and R is the correct explanation of A

(b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false

(d) A is false, but R is true

G16 TRUE / FALSE

Natural selection can produce any new trait that a population needs.

Write True or False. If false, correct the statement.

G17 TRUE / FALSE

S. aureus became resistant to penicillin in 1945, two years after the drug was first widely used.

Write True or False. If false, correct the statement.

G18 FILL IN THE BLANKS

Darwin sailed on HMS _____; the islands whose finches and tortoises influenced him most are the _____ Islands.

G19 SHORT ANSWER

Classify each as **natural** or **artificial** selection: (i) nature ‘decides’; (ii) selective breeding; (iii) breeding Dalmatians; (iv) finch beaks after a drought; (v) inbreeding occurs.

G20 SHORT ANSWER

Explain, step by step, how MRSA evolved by natural selection.

G21 SHORT ANSWER

State Darwin’s two observations and two inferences (Campbell Concept 22.2).

Section H — Population genetics and Hardy–Weinberg

Lectures 22–23 · Campbell Ch. 23 · 24 questions

H1 MCQ

The Hardy–Weinberg theorem states that, in a population's gene pool:

(a) dominant alleles increase in frequency
(b) allele and genotype frequencies remain constant from generation to generation unless acted on by agents other than sexual recombination
(c) allele frequencies always change by chance
(d) heterozygotes always disappear

H2 MCQ

In $p^2 + 2pq + q^2 = 1$, the term $2pq$ is the frequency of:

(a) homozygous recessive individuals
(b) homozygous dominant individuals
(c) heterozygous individuals
(d) the recessive allele

H3 MCQ

Which is **NOT** a condition for Hardy–Weinberg equilibrium?

(a) Random mating
(b) No mutations
(c) No gene flow
(d) Small population size

H4 MCQ

Microevolution is:

(a) a change in one individual's genes
(b) mutation in a single cell
(c) the origin of a new species
(d) a generation-to-generation change in a population's allele frequencies

H5 MCQ

Genetic drift has its greatest effect in:

(a) small populations
(b) populations with high gene flow
(c) very large populations
(d) populations in Hardy–Weinberg equilibrium

H6 MCQ

A storm carries a few birds to a remote island, where they start a new population whose allele frequencies differ from the mainland. This is:

(a) gene flow
(b) the founder effect
(c) the bottleneck effect
(d) directional selection

H7 MCQ

Gene flow (e.g. wind blowing pollen into another field) tends to:

(a) have no effect on allele frequencies
(b) increase differences between populations
(c) reduce differences between populations
(d) create new species

H8 MCQ

In evolutionary terms, an individual's fitness is:

(a) its lifespan
(b) its body size
(c) its physical strength
(d) the contribution it makes to the gene pool of the next generation

H9 MCQ

Average human birth weight is an example of:

(a) stabilizing selection
(b) genetic drift
(c) disruptive selection
(d) directional selection

H10 MCQ

In the genetic-drift example on the slide, the frequency of allele R goes $0.7 \rightarrow 0.5 \rightarrow 1.0$ over three generations. In generation 3:

(a) the population is in Hardy–Weinberg equilibrium
(b) allele r has been lost and R is fixed
(c) gene flow has added new alleles
(d) natural selection has favoured r

H11 MATCH THE FOLLOWING

Match the mode of selection with the example.

Column I	Column II
A. Directional	1. Intermediate phenotypes favoured — human birth weight
B. Disruptive (diversifying)	2. One extreme favoured — finch beak depth rising after a drought
C. Stabilizing	3. Both extremes favoured — small beaks for small seeds, large beaks for large seeds

H12 MATCH THE FOLLOWING

Match the cause of microevolution with the example.

Column I	Column II
A. Mutation	1. Wind blows pollen into another field
B. Genetic drift	2. Alleles lost by chance when only a few individuals reproduce
C. Gene flow	3. MRSA cells survive methicillin and multiply
D. Natural selection	4. A point mutation or gene duplication creates a new allele

H13 NUMERICAL

A population is in Hardy–Weinberg equilibrium and 9% of individuals show the recessive phenotype (*aa*). Find *q*, *p*, and the frequencies of *AA* and *Aa*.

H14 NUMERICAL

The next generation of pea plants has 552 plants, of which 546 have purple petals (white is recessive). Assuming Hardy–Weinberg, find the allele frequencies, and the expected number of heterozygous plants.

H15 NUMERICAL

A population of 1000 has 360 *AA*, 480 *Aa* and 160 *aa* individuals. (a) Find *p* and *q* by counting alleles. (b) Is the population in Hardy–Weinberg equilibrium?

H16 NUMERICAL

Another population of 1000 has 500 *AA*, 200 *Aa* and 300 *aa*. Find *p* and *q*, the genotype numbers expected under Hardy–Weinberg, and state whether it is in equilibrium.

H17 NUMERICAL

Phenylketonuria (PKU, recessive) affects about 1 in 10,000 newborns. Assuming Hardy–Weinberg, what percentage of the population are carriers?

H18 NUMERICAL

In a population at equilibrium, 36% of individuals show the recessive phenotype. Find the frequencies of homozygous dominant and heterozygous individuals.

H19 NUMERICAL

Scale colour in lizards: green (*G*) is dominant over blue (*g*). There are 1024 *GG*, 512 *Gg* and 64 *gg* individuals. Find the allele frequencies and the frequencies of the three genotypes. (Slide problem.)

H20 TRUE / FALSE

Gene flow tends to increase the genetic differences between populations.

Write True or False. If false, correct the statement.

H21 ASSERTION – REASON

Assertion (A): A population in Hardy–Weinberg equilibrium is not evolving.
Reason (R): In such a population, allele and genotype frequencies stay constant from generation to generation.

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

H22 ASSERTION – REASON

Assertion (A): Genetic drift has its greatest effect in very large populations.
Reason (R): Genetic drift is a change in a population's allele frequencies due to chance.

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

H23 FILL IN THE BLANKS

Fitness is the _____ an individual makes to the gene pool of the next generation. Natural selection can alter the frequency distribution of heritable traits in three ways: _____, _____ and _____ selection.

H24 SHORT ANSWER

List the five conditions for Hardy–Weinberg equilibrium and, for each, name the evolutionary force that acts when it is violated.

4 · Answers and explanations

Model paper first, then Sections A–H

Model question paper

M1 **(b) 1866** Campbell 14.1
1859 is Darwin's *Origin of Species*.

M2 **(d) *P_* × *pp*** Campbell 14.1
Crossing a dominant-phenotype individual of unknown genotype with a homozygous recessive reveals its genotype: all dominant → *PP*; 1 : 1 → *Pp*.

M3 **(a) Leaf shape** Campbell 14.1
The seven: flower colour, flower position, seed colour, seed shape, pod shape, pod colour, stem length.

M4 **(c) 92** Campbell 12.2
46 duplicated chromosomes × 2 sister chromatids = 92.

M5 **(d) prophase and prometaphase** Campbell 12.2
The centrosome replicates in interphase; the two migrate to the poles during prophase and prometaphase.

M6 **(b) non-sister chromatids of homologous chromosomes** Campbell 13.3
During prophase I; the sites are the chiasmata.

M7 **(a) 47, XXY** Campbell 15.4
Male, usually sterile.

M8 **(b) the mother is a carrier** Campbell 15.2
Sons get their X from the mother, so she must carry the recessive allele (X^HX^h).

M9 **(d) cyclic AMP** Campbell 11.3
Made from ATP by adenylyl cyclase; activates protein kinase A.

M10 **(c) how hormones act, including the second messenger cAMP** Campbell 11.1
G proteins: Gilman and Rodbell, 1994.

M11 **(a) transcriptome** Campbell 21.2

M12 **(a) left-handed**
A- and B-DNA are right-handed.

M13 **(c) hinny** Campbell 24.1
Male donkey × female horse = mule. Both are sterile.

M14 **(d) all species can produce more offspring than the environment can support** Campbell 22.2
Observation #1 is that members of a population vary in their inherited traits.

M15 **(c) 0.32** Campbell 23.2
$2pq = 2 \times 0.8 \times 0.2 = 0.32$.

M16 **(b), (c), (d)** Campbell 12.2
Nuclear envelopes re-form in telophase.

M17 **(c), (d)** Campbell 11.2
The G protein switches itself off by hydrolysing GTP (built-in GTPase), and GPCRs stay in the plasma membrane.

M18 **(a), (b), (c), (d)** Campbell 23.2
All four break a condition: gene flow, mutation, natural selection, and small size (genetic drift).

M19 **(c), (d)** Campbell 15.2
Albinism is autosomal recessive; Down syndrome is trisomy 21. (Colour blindness is the third X-linked example in the lecture.)

M20 **(d)**
Data generation has become affordable; bioinformatics began around 1960 (Dayhoff, Eck, Ledley); it covers proteins, RNA, structures, images and more.

M21 **A–3, B–5, C–1, D–2**
Griffith (transforming principle) is the distractor.

M22 **A–3, B–4, C–1, D–2** Campbell 15.4

M23 **A–3, B–4, C–2, D–1** Campbell 23.3–23.4

M24 **interphase** Campbell 12.2
G_1 + S + G_2.

M25 **furrow | plate** Campbell 12.2

M26 **False** Campbell 13.2
Haploid (*n*).

M27 **True** Campbell 15.2

M28 **hemizygous** Campbell 15.2

M29 **False** Campbell 11.3
Adenylyl cyclase makes cAMP; phosphodiesterase converts cAMP → AMP.

M30 **the study of an -ome (a complete set)** Campbell 21.2

M31 **True** Campbell 22.2

M32 **(a) 1/16 (b) 3/16** Campbell 14.2
Per gene: *Aa* × *aa* → ½ *A_*, ½ *aa*; *Bb* × *Bb* → ¾ *B_*, ¼ *bb*; *Cc* × *cc* → ½ *C_*, ½ *cc*. (a) ½ × ¼ × ½ = 1/16. (b) ½ × ¾ × ½ = 3/16.

M33 **(a) 1/2 (b) 1/2 (c) 1/2** Campbell 15.2
She must be a carrier (X^NX^n) because her father gave her X^n. × X^nY: daughters X^NX^n or X^nX^n (½ affected); sons X^NY or X^nY (½ affected).

M34 **480 heterozygous; 360 homozygous dominant** Campbell 23.2

$q^2 = 0.16 \rightarrow q = 0.4$, $p = 0.6$. $2pq = 0.48 \rightarrow 480$; $p^2 = 0.36 \rightarrow 360$.

M35 **Model answer** Campbell 11.1–11.4

Reception: adrenaline binds a GPCR. **Transduction:** the receptor activates G proteins (GDP → GTP), which activate adenylyl cyclase; it converts ATP → cAMP (second messenger); cAMP activates protein kinase A, which activates phosphorylase kinase, which activates glycogen phosphorylase. **Response:** glycogen phosphorylase breaks glycogen into glucose 1-phosphate. Each step activates many molecules of the next, giving ~10^8-fold amplification.

M36 **Model answer** Campbell 23 intro

Small, soft seeds became scarce while large, hard seeds remained. Birds with deeper beaks could crack them and survived (about 180 of ~1200) and reproduced, so average beak depth in the next generation increased. No bird's beak grew — the proportion of large-beaked birds in the population rose. So natural selection acts on individuals, but populations evolve.

Section A — Heredity and Mendel's laws

A1 **(a) Its characters vary continuously, which makes blending easy to measure** Campbell 14.1

Mendel deliberately chose characters with two distinct, 'either-or' forms (purple *or* white). The other three options are exactly the model-system criteria on the lecture slide.

A2 **(d) character; trait** Campbell 14.1

A **character** is a heritable feature that varies (flower colour); each variant of it (purple, white) is a **trait**.

A3 **(a) all purple** Campbell 14.1

No blending: every F_1 plant was purple (genotype *Pp*). White 'disappeared' in F_1 and reappeared in F_2.

A4 **(a) 3 : 1** Campbell 14.1

705 ÷ 224 = 3.15, i.e. about 3 : 1 (dominant : recessive phenotype).

A5 **(d) the blending hypothesis** Campbell 14.1

If parental 'paints' blended, white could never come back unchanged. Its return shows heredity works with discrete units (particulate hypothesis).

A6 **(b) ½ *P* : ½ *p*** Campbell 14.1

The two alleles separate during gamete formation, so each gamete carries only one of them, with equal probability.

A7 **(d) separation of homologous chromosomes during anaphase I of meiosis** Campbell 15.1

The two alleles of a gene sit at the same locus on a pair of homologous chromosomes; the homologues separate in anaphase I, taking one allele each.

A8 **(c) on different chromosomes, or far apart on the same chromosome** Campbell 14.1, 15.3

The slide's wording: inheritance of one trait does not influence another *provided the genes are on different chromosomes or far apart on the same chromosome*.

A9 **A-3, B-4, C-1, D-2** Campbell 14.1, 16.1

Timeline from the first lecture slide. 'Discovery of G proteins' (Gilman & Rodbell, Nobel 1994) is the distractor.

A10 **A-4, B-5, C-1, D-2, E-3** Campbell 14.1

F = filial (child). Definitions from the 'Terminology' slide.

A11 **gene | alleles** Campbell 14.1

e.g. the flower-colour gene has a purple allele (*P*) and a white allele (*p*).

A12 **3 : 1 | 1 : 2 : 1** Campbell 14.1

PP : *Pp* : *pp* = 1 : 2 : 1, but *PP* and *Pp* both look purple, giving 3 purple : 1 white.

A13 **genome | DNA | eye colour | height (or blood pressure)**

From the 'trait' slides (genome.gov definitions).

A14 **False** Campbell 14.1

It is hidden (recessive) in F_1 but passes on unchanged and reappears in F_2.

A15 **True** Campbell 14.1

Both are purple; one is homozygous, the other heterozygous.

A16 **False** Campbell 14.1

Segregation deals with the two alleles of **one** gene. How **different** genes behave relative to each other is independent assortment.

A17 **(a) Both A and R are true, and R is the correct explanation of A** Campbell 14.1

Both true; dominance of *P* explains why *Pp* plants are purple.

A18 **(b) Both A and R are true, but R is not the correct explanation of A** Campbell 13.4

Both are true, but R is true of every child and cannot explain why siblings differ. The real explanation: independent assortment, crossing over and random fertilization give every gamete — and so every zygote — a new combination.

A19 **(a) 200 (b) 2/3** Campbell 14.1–14.2

(a) P(*pp*) = 1/4 → 800 × 1/4 = 200. (b) Purple plants are *PP* (1/4) and *Pp* (2/4). Among purple only: (2/4) ÷ (3/4) = 2/3.

A20 **(a) 9/16 (b) 1/16 (c) 4** Campbell 14.1–14.2

(a) P(Y_) × P(R_) = 3/4 × 3/4 = 9/16. (b) 1/4 × 1/4 = 1/16. (c) *YR, Yr, yR, yr* = $2^2 = 4$.

A21 **16** Campbell 14.2

Each heterozygous gene gives 2 choices: $2^4 = 16$.

A22 **Model answer** Campbell 14.1

Each F_1 plant is *Pp* and makes ½ *P* and ½ *p* gametes. Random fertilization gives *PP* : *Pp* : *pp* = 1 : 2 : 1. Because *P* is dominant, *PP* and *Pp* are both purple, so purple : white = 3 : 1. It means heredity is carried by discrete units that come in pairs, separate into gametes, and recombine at random — not by blending.

A23 **Model answer** Campbell 14.1

In F_1 only the dominant trait shows, so the recessive factor seems to vanish. Only in F_2 does it reappear in a predictable 3 : 1 ratio. Counting thousands of F_2 plants (quantitative analysis) let him deduce segregation; stopping at F_1 the basic pattern would have eluded him.

Section B — Mendel and the gene idea: probability, dominance, pedigrees

B1 **(a) show all possible combinations of sperm and egg, to predict the results of a cross** Campbell 14.1

Capital letter = dominant allele, lowercase = recessive allele; the grid lists every egg × sperm combination.

B2 **(a) *Jj*** Campbell 14.1

Heterozygous = two different alleles (hybrid). The others are homozygous (true-breeding).

B3 **(c) One functional *P* allele makes enough active enzyme to synthesise the purple pigment** Campbell 14.1, 14.3

The white allele codes for an inactive enzyme; one working copy of *P* is sufficient for pigment. The dominant allele does not 'attack' the recessive one.

B4 **(d) locus** Campbell 14.1

Locus (plural loci) = gene location; alleles of a gene occupy the same locus on homologous chromosomes.

B5 **(b) incomplete dominance** Campbell 14.3

The heterozygote is intermediate between the parents. F_2 is 1 red : 2 pink : 1 white — phenotype ratio = genotype ratio.

B6 **(d) codominance** Campbell 14.3

Both alleles affect the phenotype in separate, distinguishable ways (both sugars present).

B7 **(c) 6 genotypes, 4 phenotypes** Campbell 14.3

Genotypes: I^AI^A, I^Ai, I^BI^B, I^Bi, I^AI^B, ii = 6. Phenotypes: A, B, AB, O = 4.

B8 **(c) one gene has multiple phenotypic effects** Campbell 14.3

Example (Campbell): the sickle-cell allele affects many characters. 'Many genes → one character' is polygenic inheritance.

B9 **(a) addition rule** Campbell 14.2

Heterozygosity can arise in two mutually exclusive ways (*R* egg + *r* sperm, or *r* egg + *R* sperm), so the probabilities are added.

B10 **(d) All of the above** Campbell 14.4

So human genetics relies on pedigrees — family trees that trace a trait across generations.

B11 **(a), (b), (d)** Campbell 14.3

'Dominant alleles are always more common' is false: dominance says nothing about how common an allele is (Campbell's example: the allele for extra fingers, polydactyly, is dominant but rare).

B12 **A–3, B–1, C–5, D–2, E–4** Campbell 14.3

Note the difference between codominance (both alleles show in one heterozygote) and multiple alleles (more than two alleles exist in the population).

B13 **A–3, B–4, C–2, D–1** Campbell 14.3

i adds neither carbohydrate, so it is recessive to both I^A and I^B.

B14 **Model answer** Campbell 14.4

(a) Recessive — two unaffected parents have an affected child, so both must carry a hidden allele. (b) Dominant — two affected parents have an unaffected child; each parent must carry a recessive normal allele that the child received from both. (c) *Aa* × *Aa*. (d) 1/4.

B15 **6/16 = 3/8** Campbell 14.2

Per gene: P(*pp*) = 1/4, P(*P_*) = 3/4; P(*yy*) = 1/2; P(*rr*) = 1/2. List the genotypes that are recessive for ≥ 2 characters:
ppyyRr 1/4 × 1/2 × 1/2 = 1/16; *ppYyrr* = 1/16; *Ppyyrr* 2/4 × 1/2 × 1/2 = 2/16; *PPyyrr* = 1/16; *ppyyrr* = 1/16.
Sum = 6/16 = 3/8.

B16 **(a) 1/64 (b) 1/8 (c) 1/64** Campbell 14.2

Treat each gene as a separate monohybrid cross and multiply. (a) 1/4 × 1/4 × 1/4. (b) 1/2 × 1/2 × 1/2. (c) 1/4 × 1/4 × 1/4.

B17 **60 red : 120 pink : 60 white** Campbell 14.3

1/4 C^RC^R (red), 1/2 C^RC^W (pink), 1/4 C^WC^W (white) × 240.

B18 **(a) 1/4 (b) 1/4 (c) 1/16** Campbell 14.2–14.3

Gametes ½ I^A, ½ i × ½ I^B, ½ i: AB, A, B, O each 1/4. Two independent births: 1/4 × 1/4.

B19 **(b) A or AB** Campbell 14.3

The mother (*ii*) can only give *i*, so the child (I^Ai) got I^A from the father — he is A or AB.

B20 **(a) 1/4 (b) 2/3 (c) 9/16** Campbell 14.4

(a) *aa* = 1/4. (b) Normal = *AA* 1/4 + *Aa* 2/4; carriers among normal = (2/4)/(3/4) = 2/3. (c) 3/4 × 3/4 = 9/16.

B21 **Model answer** Campbell 14.1, 14.3

The *P* allele codes for an active enzyme that makes purple pigment; the *p* allele's DNA sequence differs, giving an inactive enzyme. In *Pp* cells the single *P* copy makes enough active enzyme for full purple colour, so purple is dominant. The protein need not be an enzyme — any protein whose function is needed works the same way.

B22 **True** Campbell 14.4

That is a main use of pedigree analysis (and of genetic counselling).

B23 **independent | product** Campbell 14.2

Addition rule: for mutually exclusive events, add the probabilities.

Section C — Cell cycle, mitosis and meiosis

C1 **(c) 46; 23** Campbell 12.1, 13.2

Somatic cells are diploid (2*n* = 46, 23 pairs); gametes are haploid (*n* = 23).

C2 **(b) S** Campbell 12.2

S = synthesis (of DNA).

C3 **(b) Prometaphase** Campbell 12.2
Prometaphase: envelope fragments, spindle microtubules invade the nuclear area and attach to kinetochores.

C4 **(a) protein complexes associated with the centromere of each chromatid** Campbell 12.2
Spindle (kinetochore) microtubules attach here and move the chromosomes. 'Radial arrays of short microtubules' describes an aster.

C5 **(b) an imaginary plane midway between the two spindle poles** Campbell 12.2
It is not a real structure; chromosomes line up on it at metaphase.

C6 **(d) shorten by depolymerizing at their kinetochore ends** Campbell 12.2
Slide wording: "the microtubules shorten by depolymerizing at their kinetochore ends".

C7 **(d) the centrosomes, the spindle microtubules and the asters** Campbell 12.2
Spindle assembly in animal cells starts at the centrosome, which duplicates in interphase. Actin + myosin form the cleavage furrow, not the spindle.

C8 **(c) prophase I of meiosis** Campbell 13.3
Crossing over is between **non-sister** chromatids; the crossover sites are seen as chiasmata.

C9 **(a) 3 chromosomes, each of two sister chromatids** Campbell 13.3
Meiosis I separates homologues (haploid, $n = 3$), but sister chromatids stay attached until meiosis II.

C10 **(b) Separation of sister chromatids in mitosis** Campbell 13.4
Mitosis produces genetically identical cells. The other three are Campbell's three sources of variation in meiosis/sex.

C11 **(a) a cell plate formed from vesicles that fuse in the middle of the cell** Campbell 12.2
Plant cells have walls, so they build a new wall from the middle (cell plate). Animal cells pinch in (cleavage furrow).

C12 **(c), (d)** Campbell 13.3
Both divisions build a spindle, and sister chromatids separate in mitotic anaphase **and** in anaphase II. (A third meiosis-only event, not listed: homologues separating at anaphase I.)

C13 **A–4, B–2, C–5, D–1, E–3** Campbell 12.2
Order: Prophase → Prometaphase → Metaphase → Anaphase → Telophase (often followed by cytokinesis).

C14 **A–3, B–2, C–4, D–5, E–1** Campbell 12.2
Interphase = G_1 + S + G_2 (about 90% of the cycle).

C15 **(iv) Prophase → (ii) Prometaphase → (v) Metaphase → (i) Anaphase → (iii) Telophase** Campbell 12.2
Mnemonic: **P**lease **P**ass **M**e **A**nother **T**ea.

C16 **Model answer** Campbell 12.2
A – metaphase (chromosomes in a line at the plate); B – telophase / cytokinesis (two nuclei, cell pinching); C – prophase (round nucleus still intact, spindle forming outside it); D – anaphase (chromosome sets moving apart); E – prometaphase (envelope gone, condensed chromosomes scattered).

C17 **(a) 46, $2x$ (b) 46, x (c) 23, x (d) 23, $x/2$** Campbell 12.2, 13.3

S phase doubles the DNA but not the chromosome count (each chromosome now has two chromatids). Mitosis halves the DNA back to x. Meiosis I halves the chromosome number (23 duplicated chromosomes, DNA x). Meiosis II separates chromatids: 23 chromosomes, $x/2$.

C18 **(a) 16 (b) 4 (c) 16** Campbell 13.3–13.4

(a) 2^n with $n = 4$: $2^4 = 16$. (b) One tetrad per homologous pair = 4. (c) 8 chromosomes × 2 chromatids = 16.

C19 **$2^{23} \approx 8.4$ million; about 7×10^{13} (≈ 70 trillion)** Campbell 13.4

$8.4 \times 10^6 \times 8.4 \times 10^6 \approx 7.0 \times 10^{13}$. Crossing over adds even more variation.

C20 **False** Campbell 13.3

There is one round of DNA replication (before meiosis I) followed by two divisions — that is how the number is halved.

C21 **True** Campbell 12.1

(slide: “Once separate, chromatids are called chromosomes”).

C22 **False** Campbell 12.2

Mitosis divides the nucleus; cytokinesis divides the cytoplasm. Together they make up the M phase.

C23 **centromere | DNA | protein** Campbell 12.1

Chromatin condenses into visible chromosomes during cell division.

C24 **(c) A is true, but R is false** Campbell 13.3

A is true ($2n \rightarrow n$, because meiosis I separates homologues). R is false — there is no DNA replication between the two divisions.

C25 **Model answer** Campbell 13.3

Divisions: one vs two. Synapsis/crossing over: absent vs present in prophase I. Products: two diploid cells identical to the parent vs four haploid cells, genetically different from the parent and from each other. Role: growth, repair, asexual reproduction vs gamete formation and genetic diversity. (In both, DNA replicates once, during the preceding interphase.)

C26 **Model answer** Campbell 13.2

Homologous chromosomes are a matched pair — one from each parent — with the same genes at the same loci, but possibly different alleles. Sister chromatids are the two identical copies of one duplicated chromosome, made in S phase and joined at the centromere. Homologues separate in anaphase I; sister chromatids separate in mitotic anaphase and anaphase II.

Section D — Chromosomal basis of inheritance

D1 **(d) Mendelian genes have specific loci on chromosomes, and it is the chromosomes that undergo segregation and independent assortment** Campbell 15.1

The behaviour of chromosomes in meiosis explains Mendel’s laws.

D2 **(b) the eye-colour gene is on the X chromosome, with no corresponding locus on the Y** Campbell 15.1

F_1 were all red, so white is recessive; its restriction to males in F_2 shows it is X-linked.

D3 **(c) the wild-type (red-eye) allele** Campbell 15.1

+ marks the wild type, the phenotype most common in natural populations.

D4 **(b) hemizygous** Campbell 15.2

He has only one X, so only one allele of each X-linked gene.

D5 **(a) Haemophilia** Campbell 15.2

Others in the lecture: colour blindness (mostly X-linked) and Duchenne muscular dystrophy. Albinism is autosomal recessive; Down and Turner are chromosome-number disorders.

D6 **(d) $n+1$, $n+1$, $n-1$, $n-1$** Campbell 15.4

All four are abnormal. Nondisjunction of sister chromatids in meiosis II gives $n+1$, $n-1$, n, n.

D7 **(c) $n+1$, $n-1$, n, n** Campbell 15.4

Only the cell with the error makes abnormal gametes; the other meiosis II cell is normal.

D8 **(d) Karyotype X0; female; the only known viable monosomy in humans** Campbell 15.4

Turner (X0) individuals are female and usually sterile. XXY = Klinefelter (male).

D9 **(b) trisomic** Campbell 15.4

Monosomic = $2n - 1$; triploid = $3n$ (a whole extra set).

D10 **(a) on the Y chromosome; its protein triggers development of male features (testes)** Campbell 15.2

SRY = Sex-determining Region of Y.

D11 **(c) Only the ends of the Y chromosome have regions homologous with the X** Campbell 15.2

Every ovum carries X; sperm are 50% X and 50% Y. The X carries many genes unrelated to sex; the Y mainly codes for sex determination.

D12 **A–3, B–1, C–2, D–5, E–4** Campbell 15.4

Down syndrome frequency rises with the mother's age.

D13 **(a) 0 (b) all X^NX^n (carriers) (c) 0** Campbell 15.2

Daughters get the father's only X (X^n) and a normal X from the mother → carriers. Sons get the father's Y and the mother's X^N → normal.

D14 **(a) 1/2 (b) 1/2 (c) 1/4** Campbell 15.2

Children: X^NX^N, X^NX^n, X^NY, X^nY, each 1/4. Only X^nY is affected → 1/4 of all children, 1/2 of sons.

D15 **(a) 1/2, for sons and daughters alike (b) carriers (X^NX^n) (c) X^NY — they carry no recessive allele** Campbell 15.2

Children: X^NX^n, X^nX^n, X^NY, X^nY, each 1/4. Affected: X^nX^n + X^nY = 1/2.

D16 **All F_1 females red-eyed ($X^{w+}X^w$); all F_1 males white-eyed (X^wY)** Campbell 15.1–15.2

Sons receive their only X from the mother (X^w), so they are white. Daughters receive X^{w+} from the father, which is dominant.

D17 **100 white-eyed; 0 female** Campbell 15.1

F_1: $X^{w+}X^w \times X^{w+}Y$. F_2: females all red; males ½ red, ½ white. White = 1/4 of all = 100, all male.

D18 **(a) Both A and R are true, and R is the correct explanation of A** Campbell 15.2
A female needs two copies (homozygous recessive) to be affected.

D19 **(b) Both A and R are true, but R is not the correct explanation of A** Campbell 15.4
Both are true, but R does not explain A. The explanation (Campbell 15.4): one extra or missing chromosome upsets genetic balance more than an entire extra set does.

D20 **False** Campbell 15.4
It is common in plants and rare in animals.

D21 **True** Campbell 15.4

D22 **XX | XY | X | 50** Campbell 15.2
So the father's sperm decides the sex of the child.

D23 **Model answer** Campbell 15.2
The gene is on the X chromosome. A father gives his X to every daughter and his Y to every son. So all daughters receive his X^n (and, with a normal mother, become carriers), while sons receive his Y and get their X — and their colour-vision allele — from the mother.

Section E — Cell communication and signal transduction

E1 **(d) send out a chemical signal, aggregate and form fruiting bodies containing spores** Campbell 11.1
Shows that signalling is not unique to multicellular organisms.

E2 **(a) sense the local population density through the concentration of signalling molecules** Campbell 11.1
'Quorum' = the smallest number of members needed before a group can make a decision.

E3 **(b) type-α cells** Campbell 11.1
Each mating factor binds to its specific receptor on the opposite mating type, triggering responses that lead to cell fusion.

E4 **(c) an a/α diploid cell containing all the genes of both cells** Campbell 11.1
The a/α diploid can divide by mitosis or undergo meiosis to give haploids.

E5 **(a) gap junctions; plasmodesmata** Campbell 11.1
Signalling substances pass freely between neighbouring cells through them.

E6 **(c) paracrine signalling** Campbell 11.1
Para- = beside. Local regulators travel only short distances.

E7 **(d) long-distance (endocrine) signalling** Campbell 11.1
The hormone reaches every cell, but only cells with its receptor respond.

E8 **(b) reception → transduction → response** Campbell 11.1
Analogy from the slide: knock at the door (receptor, the 'discriminator') → message carried inside (the 'transducer') → action triggered (the 'amplifier').

E9 **(c) GTP** Campbell 11.2
GDP-bound = inactive; GTP-bound = active.

E10 **(d) hydrolysing its bound GTP to GDP and P_i (GTPase activity)** Campbell 11.2
This built-in control makes the response transient once the signal is gone.

E11 **(a) seven transmembrane domains** Campbell 11.2
GPCRs are extremely widespread (vision, smell, taste, development…).

E12 **(b) ATP → cAMP** Campbell 11.3
Phosphodiesterase converts cAMP → AMP, switching the second messenger off.

E13 **(a) Alfred Gilman and Martin Rodbell** Campbell 11.1, 11.3
Earl W. Sutherland won the 1971 Nobel Prize (hormone action, cAMP as second messenger).

E14 **(c) ~60%** Campbell 11.2
Example classes on the slide: β-blockers (metoprolol, atenolol), albuterol for asthma, antipsychotics acting on D_2 dopamine receptors.

E15 **(b), (c), (d)** Campbell 11.4
The three response types listed on the slide. Signalling does not rewrite the DNA sequence.

E16 **A–5, B–2, C–3, D–4, E–1** Campbell 11.1
Local: junctions, cell–cell recognition, paracrine, synaptic. Long-distance: hormones.

E17 **A–3, B–1, C–5, D–2, E–4** Campbell 11.3
First messenger outside the cell → second messenger inside.

E18 **(ii) A signalling molecule (ligand) binds the GPCR on the outside → (iv) The receptor changes shape (conformation) → (i) The activated receptor binds an inactive G protein → (v) The G protein swaps GDP for GTP, leaves the receptor and activates an enzyme → (iii) GTPase activity hydrolyses GTP; the G protein becomes inactive again** Campbell 11.2
Ligand → shape change → G protein binds → GDP/GTP exchange & enzyme activation → GTP hydrolysis (off).

E19 **(a) 10^8-fold (b) the receptor activating G proteins ($1 \rightarrow 10^2$), adenylyl cyclase making cAMP ($10^2 \rightarrow 10^4$), and glycogen phosphorylase releasing glucose 1-phosphate ($10^6 \rightarrow 10^8$) (c) 10^4** Campbell 11.4
(a) $10^8/1$. (b) Step-to-step ratios: 100, 1, 100, 1, 10, 10, 100 — three steps of 100×. (c) $10^8/10^4 = 10^4$.

E20 **(a) Both A and R are true, and R is the correct explanation of A** Campbell 11.3
The stuck-ON G protein keeps stimulating adenylyl cyclase. (Campbell 11.3.)

E21 **(d) A is false, but R is true** Campbell 11.3–11.4
A is false — the cascade releases about 10^8 molecules. R is true, and it is exactly why A is false (amplification).

E22 **False** Campbell 11.1
Bacteria (*Myxococcus*, quorum sensing) and yeast use it too. Signalling mechanisms are evolutionarily conserved.

E23 **convert energy or a message into another form | discriminator | transducer | amplifier** Campbell 11.1
From the slide ‘a knock on the door’.

E24 **Model answer** Campbell 11.3–11.4

(1) Amplification — each active molecule activates many of the next, so one hormone molecule can release ~10^8 product molecules. (2) Regulation — every step is a point where the signal can be tuned, blocked or switched off. (3) Coordination/branching — one signal can spread to several responses, and inputs from different pathways can be combined.

E25 **Model answer** Campbell 11.3

cAMP is no longer converted to AMP, so cAMP stays high after the adrenaline is gone. Protein kinase A remains active, the cascade keeps running and glycogen breakdown (glucose 1-phosphate release) is prolonged — the response is stronger and lasts longer.

Section F — Big data in biology and bioinformatics

F1 **(c) the complete set; the study of an -ome** Campbell 21.2

Genome = all the DNA; genomics = the study of genomes.

F2 **(d) the complete set of proteins of a cell or organism** Campbell 21.2

Transcriptome = RNA transcripts; metabolome = metabolites; microbiome = microbial community.

F3 **(b) Margaret Dayhoff, Richard Eck and Robert Ledley**

Their work led to the Protein Information Resource (PIR) database.

F4 **(c) 10^{18}**

kilo 10^3, mega 10^6, giga 10^9, tera 10^{12}, peta 10^{15}, exa 10^{18}, zetta 10^{21}.

F5 **(c) 10 bp per turn, 3.4 Å between base pairs, diameter 20 Å** Campbell 16.1

So one turn = 34 Å = 3.4 nm and the helix is 2 nm wide (Campbell 16.1).

F6 **(d) left-handed, with about 12 bp per turn**

Its rotation per base pair is negative (−30°), i.e. left-handed. A-DNA: right-handed, 11 bp/turn, wider (23 Å).

F7 **(b) explainable methods**

Interpretable models are intelligible by design (their parameters and architecture can be understood directly).

F8 **(a) To weigh a protein directly**

The other three reasons are on the ‘Why compare biological sequences?’ slide.

F9 **(a) mapping the sequencing reads to a known reference genome to find differences such as SNPs and indels**

Indel = insertion or deletion; SNP = single-nucleotide polymorphism.

F10 **(d) David Baker, Demis Hassabis and John Jumper for computational protein design and structure prediction (AlphaFold)**

AlphaFold predicts 3D protein structures at near-atomic accuracy.

F11 **(b) Light microscopy**

Light microscopy works from roughly cells/bacteria upward; X-ray crystallography, NMR and EM reach ~1 Å.

F12 **(a), (c), (d)**

Output: atomic-level 3D structure, with very high, independently verified accuracy.

F13 **A-3, B-5, C-1, D-2, E-4** Campbell 21.2

Other -omes on the slide: epigenome, lipidome, glycome.

F14 **A-3, B-2, C-5, D-4, E-1**

Genomics (2015 estimate): ~1 zetta-base acquired per year; 2–40 exabytes of storage.

F15 **A-3, B-1, C-2**

From the three-circle slide.

F16 **≈ 1.02 m per haploid genome; ≈ 2 m per diploid cell** Campbell 16.1

$3.0 \times 10^9 \times 3.4 \times 10^{-10}$ m = 1.02 m. Diploid = 2 × 1.02 ≈ 2.04 m — all packed into a nucleus a few µm across.

F17 **(a) 100 turns (b) 340 nm (3400 Å) (c) ≈ 91 turns** Campbell 16.1

(a) 1000 ÷ 10. (b) 1000 × 3.4 Å = 3400 Å = 340 nm. (c) 1000 ÷ 11 = 90.9.

F18 **750 MB**

$3.0 \times 10^9 \times 2$ bits = 6.0×10^9 bits = 7.5×10^8 bytes = 750 MB. Raw reads (with quality scores and repeated coverage) need far more, which is why genomics is 'big data'.

F19 **False**

They are complementary: they give different types of information and answer different questions.

F20 **True**

(with high-throughput -omics and advances in chemistry, optics, automation and computing).

F21 **single-nucleotide polymorphism | insertion | deletion | transcription factors**

ChIP = chromatin immunoprecipitation.

F22 **(b) Both A and R are true, but R is not the correct explanation of A**

Both true (PLoS Biol 2015 comparison; Dayhoff, Eck and Ledley), but R is history — the reason for today's data scale is affordable high-throughput -omics.

F23 **Model answer**

(1) The development of computational methods for studying the structure, function and evolution of genes, proteins and whole genomes. (2) The development of methods for the management and analysis of biological information arising from genomics and high-throughput experiments. (Broad slide definition: analyse biological data to extract patterns and generate hypotheses.)

F24 **Model answer**

A black-box model may predict well, but it does not tell us *why* it made a prediction or what biological pattern underlies it — and biology needs mechanisms and testable hypotheses. Explainable methods analyse a trained model afterwards (post hoc) to reveal what it learnt; interpretable models are designed so that their parameters and architecture are understandable from the start.

Section G — Evolution and natural selection

G1 **(b) 1859** Campbell 22.2

1809 = Darwin's birth year; 1866 = Mendel's paper.

G2 **(c) the unity of life, the diversity of life, and the match between organisms and their environments** Campbell 22 intro
He explained all three with descent with modification by natural selection.

G3 **(a) can interbreed in nature and produce viable, fertile offspring** Campbell 24.1
…but cannot do so with members of other such groups.

G4 **(d) horses and donkeys are separate species** Campbell 24.1
Offspring are viable but not fertile, so the parents belong to different species.

G5 **(c) organisms that reproduce only asexually (divide to multiply)** Campbell 24.1
Interbreeding is the test, so asexual organisms (and fossils) are excluded.

G6 **(b) an inherited characteristic that enhances survival and reproduction in a specific environment** Campbell 22.2
Example: finch beak shapes suited to their food.

G7 **(a) populations increase exponentially while food supply increases only linearly** Campbell 22.2
So more offspring are produced than can survive — a struggle for existence.

G8 **(d) birds with deeper beaks could crack the plentiful large, hard seeds, survived and reproduced** Campbell 23 intro
Small soft seeds were scarce; about 180 of ~1200 birds survived. The population evolved — no individual beak changed.

G9 **(b) Natural selection acts on individuals, but it is populations that evolve over time** Campbell 22.2
Key point 1 on the 'three key points' slide.

G10 **(c) 1 error in every 10^4–10^5 nucleotides added** Campbell 23.1
Many errors are repaired, but some remain — a basis of variability.

G11 **(a) inhibiting a protein that the bacteria use to build their cell walls** Campbell 22.3
MRSA strains make a different protein, so they survive methicillin and out-reproduce non-resistant strains.

G12 **A-2, B-3, C-1** Campbell 22.2
Beak shape is matched to food — an adaptation.

G13 **A-3, B-4, C-2, D-1** Campbell 22.2
Slides '1 of 4' to '4 of 4'.

G14 **(c) A is true, but R is false** Campbell 23 intro
A is true; R is false — each bird's beak stayed the same size; the population changed.

G15 **(a) Both A and R are true, and R is the correct explanation of A** Campbell 22.2
Key point 3 on the slide.

G16 **False** Campbell 22.2
It can only amplify or diminish heritable traits that already differ among individuals.

G17 **True** Campbell 22.3
Methicillin resistance followed in 1961, again two years after wide use.

G18 **Beagle | Galápagos** Campbell 22.2
He collected specimens especially from islands, i.e. isolated populations.

G19 **Model answer** Campbell 22.2
(i) natural; (ii) artificial; (iii) artificial; (iv) natural; (v) artificial. Darwin's argument: if humans can cause big changes in a short time, nature can do so over a long time.

G20 **Model answer** Campbell 22.3
S. aureus populations contain variation, including rare cells with a different cell-wall protein that methicillin cannot inhibit. When methicillin is used, susceptible cells die while resistant cells survive and reproduce, passing on the resistance gene. Over generations the resistant proportion rises, giving MRSA. The drug did not create the mutation; it selected for it.

G21 **Model answer** Campbell 22.2
Observation 1: members of a population vary in their inherited traits. Observation 2: all species can produce more offspring than the environment can support, so many fail to survive and reproduce. Inference 1: individuals whose inherited traits give a higher probability of surviving and reproducing leave more offspring. Inference 2: this unequal survival and reproduction leads to accumulation of favourable traits in the population over generations.

Section H — Population genetics and Hardy–Weinberg

H1 **(b) allele and genotype frequencies remain constant from generation to generation unless acted on by agents other than sexual recombination** Campbell 23.2
Such a population is in Hardy–Weinberg equilibrium (it is not evolving).

H2 **(c) heterozygous individuals** Campbell 23.2
p^2 = *AA*, q^2 = *aa*; p and q are allele frequencies.

H3 **(d) Small population size** Campbell 23.2
The population must be extremely large (so there is no genetic drift). The fifth condition is no natural selection.

H4 **(d) a generation-to-generation change in a population's allele frequencies** Campbell 23.2
It happens when any Hardy–Weinberg condition is not met.

H5 **(a) small populations** Campbell 23.3
Chance events change allele frequencies much more when only a few individuals reproduce.

H6 **(b) the founder effect** Campbell 23.3
Bottleneck = a drastic reduction in the size of an existing population (e.g. by a disaster).

H7 **(c) reduce differences between populations** Campbell 23.3
Populations gain or lose alleles and become more alike.

H8 **(d) the contribution it makes to the gene pool of the next generation** Campbell 23.4
Measured relative to other individuals (relative fitness).

H9 **(a) stabilizing selection** Campbell 23.4
Very low and very high birth weights have lower survival, so intermediates are favoured.

H10 **(b) allele *r* has been lost and *R* is fixed** Campbell 23.3
Only 5, then 2, of 10 plants left offspring — chance alone removed *r*.

H11 **A-2, B-3, C-1** Campbell 23.4

Draw the curves: directional shifts the peak; disruptive splits it in two; stabilizing narrows it.

H12 **A-4, B-2, C-1, D-3** Campbell 23.1–23.3

The fifth cause on the slide, sexual recombination, rearranges alleles into fresh combinations each generation.

H13 ***q* = 0.3, *p* = 0.7, *AA* = 0.49, *Aa* = 0.42** Campbell 23.2

$q^2 = 0.09 \rightarrow q = 0.3$; $p = 1 - 0.3 = 0.7$; $p^2 = 0.49$; $2pq = 2 \times 0.7 \times 0.3 = 0.42$. Check: 0.49 + 0.42 + 0.09 = 1.

H14 ***q* ≈ 0.104 (≈ 0.1), *p* ≈ 0.896 (≈ 0.9); about 103 heterozygotes** Campbell 23.2

White = 552 − 546 = 6 → $q^2 = 6/552 = 0.0109 \rightarrow q = 0.104$, $p = 0.896$. $2pq = 0.187 \rightarrow 0.187 \times 552 \approx 103$. (The slide rounds to $q \approx 0.1$, $p \approx 0.9$.)

H15 **(a) *p* = 0.6, *q* = 0.4 (b) Yes — observed 0.36 : 0.48 : 0.16 equals $p^2 : 2pq : q^2$** Campbell 23.2

A alleles = 2 × 360 + 480 = 1200 of 2000 → $p = 0.6$. Expected: $p^2 = 0.36$, $2pq = 0.48$, $q^2 = 0.16$ — same as observed.

H16 ***p* = 0.6, *q* = 0.4; expected 360 *AA* : 480 *Aa* : 160 *aa*; NOT in equilibrium** Campbell 23.2

A = 2 × 500 + 200 = 1200 of 2000 → $p = 0.6$. Same allele frequencies as before, but far fewer heterozygotes than expected (200 vs 480), so a condition (e.g. random mating) is violated.

H17 **≈ 2% ($2pq = 0.0198$, about 1 in 50)** Campbell 23.2

$q^2 = 0.0001 \rightarrow q = 0.01$; $p = 0.99$; $2pq = 2 \times 0.99 \times 0.01 = 0.0198$. (Campbell's worked example.)

H18 ***AA* = 0.16, *Aa* = 0.48** Campbell 23.2

$q^2 = 0.36 \rightarrow q = 0.6$; $p = 0.4$; $p^2 = 0.16$; $2pq = 0.48$.

H19 ***p* = 0.8, *q* = 0.2; *GG* = 0.64, *Gg* = 0.32, *gg* = 0.04** Campbell 23.2

G alleles = 2 × 1024 + 512 = 2560; $g = 512 + 2 \times 64 = 640$; total 3200. $p = 2560/3200 = 0.8$, $q = 0.2$. Observed genotype frequencies (1024/1600 etc.) equal p^2, $2pq$, q^2.

H20 **False** Campbell 23.3

Moving alleles between populations makes them more alike.

H21 **(a) Both A and R are true, and R is the correct explanation of A** Campbell 23.2

Evolution (microevolution) is defined as a change in allele frequencies.

H22 **(d) A is false, but R is true** Campbell 23.3

A is false — chance events matter most in **small** populations. R is the correct definition.

H23 **contribution | directional | disruptive (diversifying) | stabilizing** Campbell 23.4

H24 **Model answer** Campbell 23.2–23.3

Extremely large population → if small, genetic drift. No gene flow → migration causes gene flow. No mutations → mutation adds new alleles. Random mating → non-random (sexual) selection / inbreeding changes genotype frequencies. No natural selection → natural selection.